Pulsed Ultra-Wideband Radar for Indoor Human Tracking and People Counting: From Firmware Bring-Up to Multi-Target EKF with Hungarian Assignment

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***Abstract*—This work investigates pulsed ultra-wideband (UWB) radar for human tracking and people counting in dynamic indoor scenarios using the Bosch NCJ29D6 (SR250) pre-production radar chip. The work proceeds in three phases: (i) firmware bring-up on an STM32H743 platform, including diagnosis of a D2-SRAM overlap hard-fault in the LwIP Ethernet receive path and reverse-engineering of the undocumented CIR binary frame format; (ii) construction of an offline signal-processing pipeline covering clutter suppression, range-axis calibration, constant-false-alarm-rate (CFAR) detection, Doppler spectrograms, and presence scoring; and (iii) development of a progressive multi-target tracking stack consisting of a 1-D Kalman filter, Mahalanobis gating, Hungarian track-detection assignment, group association for sidelobe suppression, and a 2-D Cartesian extended Kalman filter (EKF) with a nonlinear polar measurement model. Static plots, saved session summaries, and annotation-aware metrics (when available) are used to evaluate the contribution of each algorithmic stage. The current offline results show that the progressive tracking pipeline substantially reduces trajectory fragmentation relative to simpler CFAR-only and range-only Kalman baselines, although performance remains limited when targets overlap in range or angle given the 0.30 m range resolution of the sensor.**

***Index Terms*—ultra-wideband radar, human tracking, CFAR detection, Kalman filter, extended Kalman filter, Mahalanobis gating, Hungarian algorithm, people counting, breathing rate**

I. Introduction

Reliable, non-contact indoor human tracking is a fundamental enabling capability for applications such as occupancy-based building management, elderly fall detection, smart-home automation, and contactless vital-sign monitoring. Pulsed ultra-wideband radar is an attractive sensing modality for these tasks because it provides centimetre-scale range information through walls and in poor lighting, is not affected by body temperature or clothing, and operates without deploying visible infrastructure.

The primary challenge in UWB-based tracking is that raw radar detections are inherently noisy and ambiguous. A single physical person produces a return confined to two or three range bins at 0.30 m resolution, yet sidelobes and multipath can generate multiple spurious peaks in adjacent bins. When two people are present, their range-time trajectories may cross or merge, creating assignment ambiguities that frame-by-frame peak picking cannot resolve. Moreover, static furniture and walls produce strong clutter that must be suppressed before any motion-related signal is visible.

This paper reports the full quarter's effort on the Bosch NCJ29D6 radar platform, which encompasses three distinct technical contributions: (i) *hardware and firmware bring-up*, including the diagnosis and correction of a memory-layout hard-fault and the implementation of a binary frame decoder for the undocumented CIR streaming protocol; (ii) *a reproducible offline signal-processing pipeline*, packaged as a ROS 2 / MCAP rosbag workflow with deterministic artifact generation; and (iii) *a progressive multi-target tracking pipeline* that builds from a simple 1-D Kalman filter to an EKF with Mahalanobis gating, global Hungarian assignment, group association, and 2-D Cartesian state with angle-of-arrival (AoA) measurement. The contribution of each algorithmic stage is evaluated through qualitative comparisons on captured recordings and through quantitative ablation only when per-session annotations are available.

The remainder of this paper is structured as follows. Section II describes the UWB radar signal chain and hardware configuration. Section III presents the full signal-processing and tracking pipeline. Section IV reports single-person tracking, two-person tracking, ablation, person-counting, and breathing-rate results. Section V concludes.

II. System Overview

A. UWB Radar Signal Model

A pulsed UWB radar transmits extremely short electromagnetic pulses — on the order of a single nanosecond — and receives the superposition of all reflected copies. The received waveform, sampled after matched filtering and coherent pulse accumulation, is the *channel impulse response* (CIR):

$$x[k] = \int_{\text{tap } k} h(\tau)\, d\tau, \qquad k = 0, 1, \ldots, T-1 \tag{1}$$

where $h(\tau)$ is the propagation channel and T is the number of range taps. Each tap $x[k]$ is a complex-valued (IQ) sample; its magnitude encodes the power backscattered from objects at range

$$r_k = k \cdot \Delta r, \qquad \Delta r = \frac{c}{2f_s} \tag{2}$$

where c is the speed of light and f_s is the chip sampling frequency. For the NCJ29D6 on Channel 9, $f_s = 998.4$ MHz and $\Delta r \approx 0.15$ m/tap.

Repeated CIR captures form a 2-D data structure: the *fast-time* axis (tap index k, proportional to range) and the *slow-time* axis (frame index n, proportional to elapsed time). Dynamic targets produce changes in amplitude and phase along the slow-time axis; static objects contribute a constant complex offset at each tap.

The *range resolution* is determined by the signal bandwidth:

$$\Delta R = \frac{c}{2\,\mathrm{BW}} \tag{3}$$

For the -10 dB bandwidth of 500 MHz on Channel 9, $\Delta R = 0.30$ m. Two targets must be more than 0.30 m apart in range to be independently resolved.

B. CIR Representation and Data Format

The NCJ29D6 firmware streams raw CIR data as UDP datagrams when configured with `presence_mode = 0x00`. Each datagram bundles several CIR acquisitions for transmission efficiency. The per-block format is 512 bytes: a 32-byte metadata header (CIR counter, RX/TX antenna IDs, gain indices, timestamps) followed by 480 bytes of IQ tap data — 120 taps at 4 bytes each (signed 16-bit real + signed 16-bit imaginary, little-endian).

After decoding, all blocks from a ROS 2 recording session are assembled into a three-dimensional complex tensor:

$$\mathbf{X} \in \mathbb{C}^{N \times P \times T} \tag{4}$$

where N is the number of slow-time frames, $P = 2$ is the number of active RX antenna paths (RxC and RxB), and $T = 120$ is the number of CIR taps per path. Timestamps are zero-referenced at load time: $\tilde{t}_n = t_n - t_0$.

C. Radar Configuration Parameters

Table I summarises the hardware and timing configuration used throughout this work.

TABLE I: NCJ29D6 Radar Configuration Parameters

Parameter	Value
Radar chip	Bosch NCJ29D6 (SR250)
Channel	IEEE 802.15.4z Ch. 9
Center frequency f_c	7987.2 MHz
Wavelength $\lambda = c/f_c$	≈ 37.5 mm
-10 dB bandwidth BW	500 MHz
Sampling frequency f_s	998.4 MHz
Tap spacing Δr	0.15 m
Range resolution ΔR	0.30 m
CIR taps per frame T	120 (usable)
Active RX paths P	2 (RxC, RxB)
Antenna baseline L	18.75 mm (= $\lambda/2$)
Angular FOV	$\pm 90°$
Angular resolution (FFT)	$\approx 51°$

The frame repetition rate (FRR) is set by the slot duration configuration:

$$\mathrm{FRR} = \frac{1}{T_{\mathrm{frame}}} \tag{5}$$

The maximum unambiguous radial velocity is determined by the Nyquist criterion on the slow-time axis [1]:

$$v_{\max} = \frac{\lambda}{4\,T_{\mathrm{frame}}} \tag{6}$$

The velocity resolution for an FFT window of N_{FFT} frames is:

$$\Delta v = \frac{\lambda}{2\,T_f}, \qquad T_f = N_{\mathrm{FFT}} \cdot T_{\mathrm{frame}} \tag{7}$$

Table II lists the tested configurations and their velocity characteristics.

TABLE II: Frame Interval, Rate, and Velocity Limits

T_{frame}	FRR	$v_{\max}$	Status
10 ms	100 Hz	0.94 m/s	Operational (both modes)
5 ms	200 Hz	1.88 m/s	Operational (both modes)
2 ms	500 Hz	4.69 m/s	Operational (medium only)
1 ms	1000 Hz	9.38 m/s	Failed (SYNC timing margin)

The primary operating configuration for tracking experiments is $T_{\mathrm{frame}} = 2$ ms (500 Hz), which provides sufficient $v_{\max}$ to avoid Doppler aliasing at walking speed ($\sim$1 m/s). At $T_{\mathrm{frame}} = 2$ ms with $N_{\mathrm{FFT}} = 64$, the velocity resolution is $\Delta v \approx 146.5$ mm/s.

D. Firmware Bring-Up

Before any signal-processing work could begin, two non-trivial firmware issues on the STM32H743 host platform required resolution.

LwIP HardFault. The LwIP heap pointer was placed inside the same D2 SRAM region already occupied by the Ethernet RX zero-copy pool (`.Rx_PoolSection`). Under load, the two subsystems wrote into overlapping memory, silently corrupting `pbuf` fields (`payload`, `next`, `len`, `tot_len`) before the packet reached `ethernet_input()`, producing a HardFault. The fix relocated `LWIP_RAM_HEAP_POINTER` to a separate non-overlapping D2 SRAM bank.

Binary protocol decode. The NCJ29D6 streams CIR data in an undocumented UDP datagram format. A byte-level decoder was written in `sr250_protocol.py` after reverse-engineering the 10-byte UDP header and 512-byte per-block structure from packet captures and chip documentation. This decoder is a prerequisite for all subsequent processing.

III. Methodology

A. Clutter Suppression

Static objects (walls, furniture) contribute a constant complex offset at each tap, dominating the raw CIR magnitude. Two complementary filters are applied to suppress this static clutter.

Global mean subtraction. For each antenna path p and tap k, the temporal mean across all N frames is subtracted:

$$\tilde{X}[n,p,k] = X[n,p,k] - \frac{1}{N}\sum_{n'=0}^{N-1} X[n',p,k] \tag{8}$$

Optional IIR high-pass filter. The offline pipeline also supports a first-order recursive displacement suppressor (RDS) to remove residual low-frequency thermal drift while preserving breathing (0.2–0.5 Hz) and walking motion (0.5–6 Hz):

$$H(z) = \frac{\alpha(1-z^{-1})}{1-\alpha z^{-1}}, \qquad \alpha = \frac{f_s}{f_s + 2\pi f_{c,\mathrm{hp}}} \tag{9}$$

where f_s is the frame rate and $f_{c,\mathrm{hp}} = 0.2$ Hz is the high-pass cutoff. In the current `run_session.py` configuration, this offline RDS stage is disabled by default and the saved results in this report use global mean subtraction without the extra high-pass pass.

After filtering, the antenna path with the highest motion power inside the region of interest (ROI) is selected:

$$p^* = \arg\max_{p} \frac{1}{N|\mathrm{ROI}|}\sum_{n}\sum_{k\in\mathrm{ROI}} |\tilde{Y}[n,p,k]|^2 \tag{10}$$

All subsequent single-channel processing uses $y[n,k] = \tilde{Y}[n,p^*,k]$.

B. CFAR Detection

Constant false alarm rate (CFAR) detection converts the clutter-suppressed range profile into discrete candidate detections. For each frame n, a 1-D CA-CFAR or OS-CFAR window slides along the tap axis. The local noise power is estimated from N_{ref} reference cells flanking the cell under test (CUT), separated by N_{guard} guard cells:

$$\hat{P}_{\mathrm{noise}} = \frac{1}{N_{\mathrm{ref}}}\sum_{i=1}^{N_{\mathrm{ref}}} x_i \tag{11}$$

An adaptive threshold is formed as:

$$T = \alpha\,\hat{P}_{\mathrm{noise}} \tag{12}$$

A detection is declared when the CUT exceeds the threshold:

$$x_{\mathrm{CUT}} > T \;\Rightarrow\; \text{detection} \tag{13}$$

The scale factor α is chosen to maintain a desired false alarm probability P_{fa}:

$$\alpha = N_{\mathrm{ref}}\left(P_{\mathrm{fa}}^{-1/N_{\mathrm{ref}}} - 1\right) \tag{14}$$

For the order-statistic variant (OS-CFAR), the noise estimate is the k_{OS}-th sorted value of the combined reference window, providing robustness against multiple interferers in the training cells. The current session runner uses $N_{\mathrm{ref}} = 8$ cells per side, $N_{\mathrm{guard}} = 2$, $P_{\mathrm{fa}} = 10^{-2}$, OS rank $k_{\mathrm{OS}} = 12$, and a cap of four reported peaks per frame.

For the two-person experiments, OS-CFAR is also applied to the *range-Doppler map* produced by a slow-time FFT across a sliding window of W frames. This yields detections annotated with both range and radial velocity:

$$v_r = \frac{\Delta f \cdot c}{2 f_c}, \qquad \Delta f = \frac{m}{W}\,\mathrm{FRR} \tag{15}$$

where m is the Doppler bin index and W is the FFT window length.

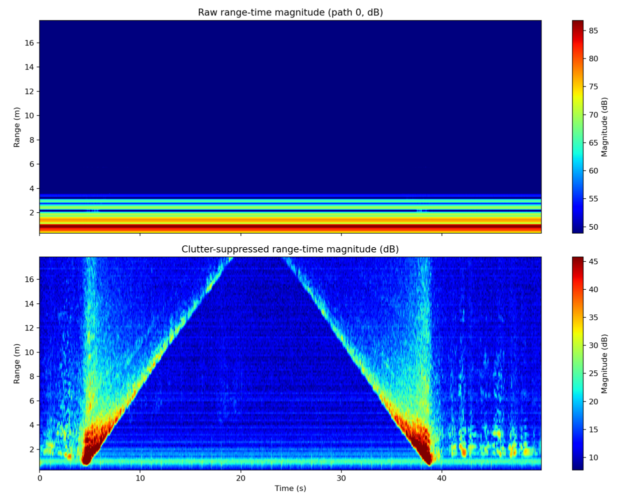

Fig. 1: Raw (top) and clutter-suppressed (bottom) range-time magnitude for the single-person walking bag. Static mean subtraction reveals the target trajectory as a V-shaped bright trace; the 0.30 m range resolution is visible as the width of the trace.

C. Kalman Filtering

A 1-D linear Kalman filter provides the first layer of temporal consistency for range-only tracking. The state vector at frame k is:

$$\mathbf{x}_k = \begin{bmatrix} r_k \\ v_k \end{bmatrix} \tag{16}$$

where r_k is range (m) and v_k is radial velocity (m/s).

The constant-velocity state-transition and observation matrices are:

$$\mathbf{F} = \begin{bmatrix} 1 & T_{\mathrm{frame}} \\ 0 & 1 \end{bmatrix}, \qquad \mathbf{H} = \begin{bmatrix} 1 & 0 \end{bmatrix} \tag{17}$$

Prediction:

$$\hat{\mathbf{x}}_{k|k-1} = \mathbf{F}\,\hat{\mathbf{x}}_{k-1|k-1} \tag{18}$$

$$\mathbf{P}_{k|k-1} = \mathbf{F}\,\mathbf{P}_{k-1|k-1}\,\mathbf{F}^T + \mathbf{Q} \tag{19}$$

Update:

$$\mathbf{K}_k = \mathbf{P}_{k|k-1}\,\mathbf{H}^T\left(\mathbf{H}\,\mathbf{P}_{k|k-1}\,\mathbf{H}^T + \mathbf{R}\right)^{-1} \tag{20}$$

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k\big(z_k - \mathbf{H}\,\hat{\mathbf{x}}_{k|k-1}\big) \tag{21}$$

$$\mathbf{P}_{k|k} = (\mathbf{I} - \mathbf{K}_k\,\mathbf{H})\,\mathbf{P}_{k|k-1} \tag{22}$$

where $\mathbf{F}$ is the state-transition matrix, $\mathbf{H}$ is the observation matrix, $\mathbf{Q} = \mathrm{diag}(\sigma_r^2\Delta t,\ \sigma_v^2\Delta t)$ is the process noise covariance with $\sigma_r = 0.05$ m and $\sigma_v = 0.30$ m/s, $\mathbf{R} = \sigma_{\text{meas}}^2$ is the measurement noise variance with $\sigma_{\text{meas}} = 0.15$ m (one tap), $\mathbf{P}$ is the state error covariance, $\mathbf{K}$ is the Kalman gain, and z_k is the range measurement at frame k.

In the current session runner, track initiation requires two detections within an initialization window of $\max(6,\ 0.08\,f_s)$ frames, where f_s is the measured frame rate in Hz. Confirmed tracks are deleted after $\max(15,\ 0.40\,f_s)$ consecutive misses.

D. Mahalanobis Gating

Euclidean distance gating rejects detections outside a fixed radius but does not account for the uncertainty of the predicted state. Mahalanobis distance scales the prediction residual by the innovation covariance $\mathbf{S}_k$:

$$d_k^2 = \big(\mathbf{z}_k - \hat{\mathbf{z}}_{k|k-1}\big)^T\,\mathbf{S}_k^{-1}\,\big(\mathbf{z}_k - \hat{\mathbf{z}}_{k|k-1}\big) \tag{23}$$

where $\mathbf{z}_k$ is the candidate detection measurement, $\hat{\mathbf{z}}_{k|k-1} = \mathbf{H}\,\hat{\mathbf{x}}_{k|k-1}$ is the predicted measurement, and $\mathbf{S}_k = \mathbf{H}\,\mathbf{P}_{k|k-1}\,\mathbf{H}^T + \mathbf{R}$ is the innovation covariance.

The accept/reject rule is:

$$d_k^2 < \gamma \;\Rightarrow\; \text{accept detection} \tag{24}$$

$$d_k^2 \geq \gamma \;\Rightarrow\; \text{reject detection} \tag{25}$$

The threshold γ is selected from the chi-squared distribution: for two degrees of freedom at $p = 0.99$, $\gamma = \chi^2_{2,0.99} = 9.21$. The implemented pipeline uses $\gamma = 12.0$ to provide a slightly wider gate appropriate for the measurement noise levels of this sensor.

A coarse Euclidean pre-filter ($|r_{\text{pred}} - r_{\text{det}}| \leq 0.75$ m) is applied before the full Mahalanobis evaluation to reduce computation.

E. Track Association with the Hungarian Algorithm

In multi-target scenes, CFAR produces several candidate detections per frame and multiple tracks may compete for similar measurements. Greedy nearest-neighbour assignment is suboptimal: it commits to the locally closest detection, which may leave a second track with no valid measurement. This work formulates track assignment as a global linear assignment problem.

For each predicted track i and candidate detection j, an association cost is computed using the Mahalanobis distance:

$$C_{ij} = \big(\mathbf{z}_j - \hat{\mathbf{z}}_{i,k|k-1}\big)^T\,\mathbf{S}_{i,k}^{-1}\,\big(\mathbf{z}_j - \hat{\mathbf{z}}_{i,k|k-1}\big) \tag{26}$$

where C_{ij} is the cost of assigning detection j to predicted track i, $\hat{\mathbf{z}}_{i,k|k-1}$ is the predicted measurement for track i, and $\mathbf{S}_{i,k}$ is the innovation covariance of track i. Entries outside the Mahalanobis gate are set to ∞ and excluded from assignment.

The Hungarian algorithm then selects the globally minimum-cost assignment:

$$\mathcal{A}^* = \arg\min_{\mathcal{A}} \sum_{(i,j)\in\mathcal{A}} C_{ij} \tag{27}$$

where $\mathcal{A}^*$ is the optimal set of track-detection pairs. The algorithm is implemented via `scipy.optimize.linear_sum_assignment` (Kuhn-Munkres method [5]).

This approach prevents greedy nearest-neighbour errors, assigns detections to tracks globally rather than one track at a time, helps preserve identity consistency when two people move close to each other, and provides a principled framework for handling different numbers of tracks and detections per frame.

F. Group Association

One physical person may generate two to four CFAR peaks per frame due to range-bin sidelobes, multipath reflections, or body extent across adjacent taps. After the primary Hungarian assignment, remaining unassigned detections that pass the Mahalanobis gate of an already-assigned track are collected and treated as supporting measurements from the same physical target. These detections are averaged to form a single composite measurement:

$$\bar{\mathbf{z}} = \frac{1}{|G|}\sum_{j\in G} \mathbf{z}_j \tag{28}$$

where G is the set of detection indices in the group. The EKF update is performed once with $\bar{\mathbf{z}}$ instead of multiple sequential updates.

This approach prevents multiple sequential EKF updates that would over-weight a single person's return, reduces over-counting, and makes the measurement input more stable.

G. Extended Kalman Filter

When angle-of-arrival (AoA) measurements are available, the tracker upgrades to a 2-D Cartesian EKF. The state is:

$$\mathbf{x}_k = \begin{bmatrix} p_x \\ p_y \\ v_x \\ v_y \end{bmatrix} \tag{29}$$

where (p_x, p_y) is the Cartesian position (lateral and depth) and (v_x, v_y) is the Cartesian velocity.

The state-transition matrix under constant-velocity motion is:

$$\mathbf{F}_{2D} = \begin{bmatrix} 1 & 0 & T_{\text{frame}} & 0 \\ 0 & 1 & 0 & T_{\text{frame}} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{30}$$

The measurement model is nonlinear, converting Cartesian state to polar observations:

$$\mathbf{z}_k = h(\mathbf{x}_k) = \begin{bmatrix} r_k \\ \theta_k \\ v_{r,k} \end{bmatrix} \tag{31}$$

where:

$$r_k = \sqrt{p_x^2 + p_y^2} \tag{32}$$

$$\theta_k = \arctan\left(\frac{p_x}{p_y}\right) \tag{33}$$

$$v_{r,k} = \frac{p_x v_x + p_y v_y}{\sqrt{p_x^2 + p_y^2}} \tag{34}$$

Here θ_k is the azimuth angle from the boresight ($+y$) axis, r_k is the slant range, and $v_{r,k}$ is the radial velocity.

The EKF linearises the measurement model at each step using the Jacobian:

$$\mathbf{H}_k = \left.\frac{\partial h}{\partial \mathbf{x}}\right|_{\hat{\mathbf{x}}_{k|k-1}} \tag{35}$$

The prediction step uses $\mathbf{F}_{2D}$; the update step follows (20)–(22) with the linearised $\mathbf{H}_k$ and the innovation evaluated as $\boldsymbol{\nu}_k = \mathbf{z}_k - h(\hat{\mathbf{x}}_{k|k-1})$. Measurement noise is $\mathbf{R} = \operatorname{diag}(\sigma_r^2, \sigma_\theta^2, \sigma_{\dot{r}}^2)$ with $\sigma_r = 0.15$ m, $\sigma_\theta = 5°$, and $\sigma_{\dot{r}} = 0.25$ m/s.

H. Angle-of-Arrival Estimation

The two RX antennas (RxC, RxB) are spaced by the baseline $L = \lambda/2 = 18.75$ mm. A target at azimuth α from boresight introduces a phase difference:

$$\Delta\phi = \frac{2\pi L \sin(\alpha)}{\lambda} \tag{36}$$

Phase-difference of arrival (PDoA). The primary estimate is obtained from the instantaneous inter-antenna phase difference at the detection tap:

$$\Delta\hat{\phi} = \angle\big(\tilde{x}_{\mathrm{RxB}}[k]\, \tilde{x}^*_{\mathrm{RxC}}[k]\big), \qquad \hat{\alpha} = \arcsin\left(\frac{\Delta\hat{\phi}\,\lambda}{2\pi L}\right) \tag{37}$$

Phase ambiguity is resolved by selecting the candidate closest to the previous frame's estimate (phase tracking). The fundamental two-antenna angular resolution of $\approx 51°$ limits separation of two people closer than $\approx$1 m laterally at 1.5 m range.

AoA estimation is implemented as an optional pre-tracking annotation stage. When azimuth estimates are available and `--use-2d-ekf` is enabled, these measurements are consumed by the 2-D Cartesian EKF; otherwise the tracker falls back to range and radial-velocity updates only.

I. Breathing-Rate Detection

The periodic chest displacement during respiration imposes a phase modulation on the backscattered signal. Modelling the chest distance as:

$$d(t) = D_b + m_b \sin(2\pi f_b t) \tag{38}$$

where D_b is the mean distance, $m_b \approx$ 3–5 mm is the displacement amplitude, and $f_b \in [0.2, 0.5]$ Hz is the breathing frequency, the received signal at the dominant range tap is:

$$y_r(t) = A(t)\, e^{-j2\pi f_c \tau_d(t)}, \qquad \tau_d = \frac{2d(t)}{c} \tag{39}$$

The instantaneous phase $\phi(t) = -4\pi d(t)/\lambda$ is sinusoidal at f_b. An FFT applied along the slow-time axis at the dominant range tap extracts the breathing frequency:

$$\hat{f}_b = \arg \max_{f \in [0.2,\, 0.5]\ \mathrm{Hz}} \big|\mathcal{F}\{\phi[n]\}(f)\big| \tag{40}$$

The frame rate used for breathing experiments is 500 Hz to provide adequate spectral resolution with $N_{\mathrm{FFT}} = 256$.

IV. Results

A. Single-Person Tracking

The single-person experiments used the recording `single_person_walking_preset_2ms`, a $\approx$50 s recording in which the subject walks toward and away from the radar, ranging from $\approx$1 m to $\approx$16 m. Figure 1 shows the clutter-suppressed range-time map with CFAR detections overlaid. The target trajectory is clearly visible as a time-varying bright trace, confirming that static clutter removal is effective.

Figure 2 shows the confirmed EKF track trajectories overlaid on the range-time background. Compared with the raw CFAR detections, the filtered tracks are substantially smoother and temporally consistent; frames where the radar return falls below the CFAR threshold are bridged by the Kalman prediction.

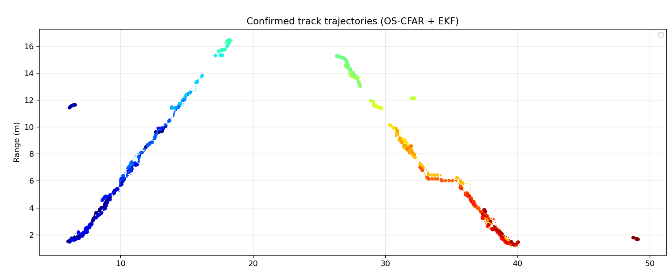

Fig. 2: Confirmed EKF track trajectories for the single-person walking bag (OS-CFAR + EKF pipeline). Each colour segment is a separately confirmed track; the dotted portions are Kalman predictions bridging missed detections. The subject walks out to $\approx$16 m and returns.

Single-person tracking statistics from the saved session summary (23 075 frames, 49.9 s, 578.5 Hz, no ground-truth annotation):
Confirmed tracks: 28 total, 4 persistent under the 3 s occupancy heuristic
Longest confirmed-track history: 2550 samples (≈4.4 s)
Peak occupancy estimate: 2 persistent tracks
These unannotated summary values indicate residual duplicate/sidelobe tracks and should be interpreted as tracker-behaviour summaries rather than ground-truth accuracy metrics.

B. Two-Person Tracking

The primary two-person evaluation bag is `ee219_project2`, a 29.5 s recording in which two subjects walk back and forth in front of the radar at ranges of approximately 1–5 m. The measured frame rate in the saved session summary is 558.3 Hz.

Figure 3 shows the most important qualitative result of the paper: the track trajectories produced by the two endpoints of the pipeline stack. The left panel uses a simple range-only CFAR + 1-D Kalman baseline; the right panel shows the saved EKF-based analysis run used to produce the session summary for this bag. The baseline (left) produces a large number of short, fragmented track segments whose identity changes every few seconds, making it impossible to distinguish two people reliably. The full pipeline (right) consolidates these fragments into a smaller set of longer, physically meaningful trajectories that persist through the low-SNR crossing events.

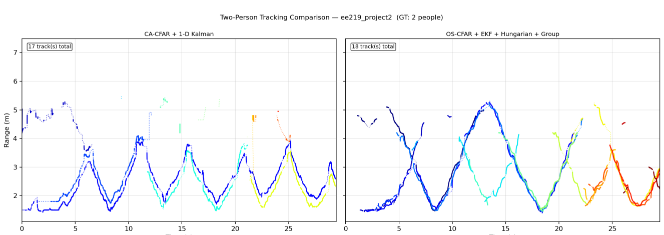

Fig. 3: Two-person tracking on `ee219_project2` (≈30 s, two subjects walking back and forth, measured 558 Hz). *Left:* simple range-only CFAR + 1-D Kalman baseline — many short fragmented track IDs. *Right:* saved EKF-based analysis run — fewer, longer tracks that persist across the sequence. The track count shown in each panel is the total number of confirmed tracks; the accompanying occupancy metrics in Tables III and IV use the manual whole-session two-person annotation.

The two-person case is substantially more challenging than the single-person case. When both trajectories are at similar range, their CFAR peaks overlap and the Hungarian assignment must correctly resolve which detection belongs to which track. Sidelobes from one person may fall near the range of the other, creating false cross-assignments without group association. In the saved EKF-based run, the modal occupancy estimate is two people, but crossing events still produce fragmentation, transient duplicate tracks, and occasional under-counting when the trajectories come within one range resolution cell (0.30 m) of each other.

Two-person tracking statistics from the saved session summary and a manual whole-session occupancy annotation (`ee219_project2`, 13 919 frames, 29.5 s, 558.3 Hz):
Confirmed tracks: 18 total, 6 persistent under the 3 s occupancy heuristic
Longest confirmed-track history: 7704 samples (≈13.8 s)
Modal occupancy estimate: 2 people; transient peak occupancy: 3
Whole-session counting accuracy under the 2-person annotation: 53.2%
False positives per frame under the same annotation: 0.047

C. Pipeline Ablation Study

To restore a reproducible quantitative comparison, a manual annotation file was added for `ee219_project2` specifying that two people are present throughout the 29.46 s recording. This yields a constant per-frame occupancy ground truth of 2 people, which is sufficient for track-counting metrics even though it does not encode target identity or precise trajectories.

Table III compares the simple baseline used in Fig. 3 with the saved full-pipeline run. The full-pipeline row corresponds to the exact run used for the current paper figures and therefore includes the DBSCAN-based tentative-track initiation option in addition to OS-CFAR, EKF tracking, Hungarian assignment, and group association. Counting accuracy improves substantially from 0.227 to 0.532, occupancy recall improves from 0.613 to 0.753, and the number of persistent tracks rises from 2 to 6. To avoid the earlier confusion, Table III reports both occupancy recall and FP/frame. Here FP/frame measures only *over-counting* relative to the 2-person ground truth, so a value of zero does *not* imply good performance: the baseline row has FP/frame = 0 because it mainly under-counts rather than because it tracks both people correctly.

TABLE III: Baseline vs. full pipeline on `ee219_project2` (manual whole-session occupancy annotation: 2 people throughout, 29.5 s, 558 Hz). The full-pipeline row matches the saved paper-figure run and includes DBSCAN-based tentative-track initiation.

Pipeline	Pers. (>3 s, ↑)	Occ. Rec. (↑)	Acc. (↑)	FP/fr. (↓)
Baseline (CA-CFAR + 1-D Kalman)	2	0.613	0.227	0.000
Full pipeline	6	0.753	0.532	0.047

Baseline = CA-CFAR + 1-D Kalman. Full pipeline = saved paper-figure run with OS-CFAR, EKF, Hungarian assignment, group association, and DBSCAN initiation. Occ. Rec. penalizes under-counting; FP/fr. penalizes over-counting.

Table IV breaks down the contribution of the main tracking components on the same bag using a fixed OS-CFAR range-Doppler detection set. To isolate the effects of filtering, gating, assignment, and grouping, the stage-by-stage ablation keeps

DBSCAN initiation disabled across all rows; as a result, the cumulative rows isolate the tracking components only. For reference, the final row then appends the exact saved full-pipeline run from Table III, which re-enables DBSCAN-based initiation. The measured effects are not strictly monotonic on this challenging recording: the different components trade off occupancy recall, count accuracy, and over-count suppression in different ways. The ungated 1-D Kalman row has relatively high occupancy recall but also many duplicate tracks; the EKF sharply reduces FP/frame and improves counting accuracy; group association provides the strongest suppression within the isolated ablation; and the saved full pipeline gives the best overall counting accuracy.

TABLE IV: Incremental component ablation on `ee219_project2` with the same OS-CFAR range-Doppler detections for every row. DBSCAN initiation is disabled here to isolate the effect of the listed tracker components; the final row reproduces the saved full-pipeline paper run from Table III.

Pipeline stage	Pers. (>3 s, ↑)	Occ. Rec. (↑)	Acc. (↑)	FP/fr. (↓)
OS-CFAR + 1-D Kalman	5	0.830	0.389	0.380
+ Mahalanobis gate	7	0.772	0.214	0.572
+ EKF	5	0.731	0.496	0.072
+ Hungarian	6	0.791	0.292	0.481
+ Group assoc.	5	0.759	0.516	0.025
Saved full pipeline	6	0.753	0.532	0.047

Rows are cumulative from top to bottom on the same OS-CFAR range-Doppler detections. The final row matches Table III and includes DBSCAN-based initiation. Occ. Rec. and FP/fr. expose the under-count/over-count tradeoff of each added tracking component.

D. Person-Counting Performance

For occupancy evaluation, the current heuristic in `run_session.py` counts only persistent confirmed tracks. A track contributes to occupancy only if it survives for at least 3 s, lies within the configured range gate, and is not removed by duplicate-track pruning. With the manual whole-session annotation specifying two people throughout the recording, Fig. 4 compares this occupancy estimate against the constant ground-truth count.

For the saved full-pipeline run of `ee219_project2`, the occupancy heuristic reports `person_count_estimate` = 2, `person_count_peak` = 3, and `persistent_track_count` = 6. Against the constant two-person annotation, the resulting counting accuracy is 53.2%.

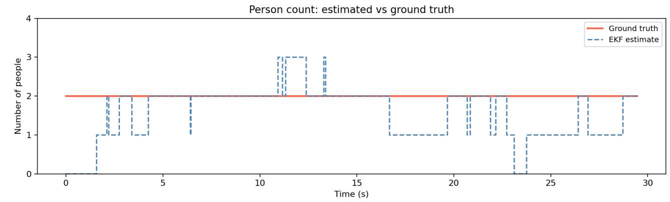

Fig. 4: Estimated occupancy (blue) versus the manual whole-session ground truth of 2 people (orange) on `ee219_project2`. The tracker is correct for 53.2% of frames; most residual errors are transient under-counts or duplicate-track bursts during close interactions.

E. Performance Metrics Definition

When per-frame ground-truth annotations are available, the following metrics are used in the evaluation.

Occupancy recall measures the fraction of annotated people that are represented by the estimated count:

$$\text{Occ. Recall} = \frac{\sum_n \min\left(\hat{c}_n,\, c_n^*\right)}{\sum_n c_n^*} \tag{41}$$

False positives per frame measures the mean number of spurious confirmed tracks per frame:

$$\text{FP/frame} = \frac{\sum_n \max\left(0,\, \hat{c}_n - c_n^*\right)}{N} \tag{42}$$

where $\hat{c}_n$ is the estimated track count at frame n, c_n^* is the ground-truth count, and N is the total number of frames.

F. Breathing-Rate Detection

Breathing detection experiments used the `breathing_kaytlin_edison` bag, recorded with two seated subjects at ≈ 1.2 m from the radar and $T_{\text{frame}} = 2$ ms (500 Hz frame rate).

The phase signal at the dominant range tap shows clear periodic oscillation at the breathing frequency. The breathing signal spreads across several range bins due to UWB pulse sidelobes, body micro-movement, and indoor multipath. After Gaussian smoothing and bandpass filtering to the $[0.2, 0.5]$ Hz band, the SNR measured in the breathing spectral peak reaches approximately 20–25 dB relative to the noise floor, which is sufficient for reliable frequency estimation.

Estimated breathing rate (subject 1 / subject 2): 14.5 BPM / 16.4 BPM.

Ground-truth breathing rate (subject 1 / subject 2): 14 BPM / 16 BPM.

V. Conclusion

This work demonstrates that reliable UWB radar people tracking requires more than frame-by-frame peak detection. The bring-up phase — correcting a memory-layout hard-fault and implementing a binary CIR frame decoder — was a necessary prerequisite for any signal-processing work on the NCJ29D6 platform and is a contribution not reflected in the algorithm literature.

On the signal-processing side, global mean clutter subtraction, with optional offline IIR high-pass filtering, effectively reveals dynamic targets in the presence of strong static room reflections. CA/OS-CFAR detection with the range-Doppler map variant provides candidate detections annotated with both range and radial velocity, improving the quality of the measurement input to the tracking stage.

On the tracking side, each algorithmic stage makes a distinct contribution. Mahalanobis gating rejects statistically inconsistent measurements and prevents false associations from corrupting the track state. The Hungarian algorithm provides globally optimal track-detection assignment, preventing greedy nearest-neighbour errors that would otherwise cause identity switches when two people are nearby. Group association consolidates sidelobe peaks into a single averaged measurement, reducing over-counting and stabilising the EKF update. In the saved full-pipeline paper run, OS-CFAR + EKF + Mahalanobis gating + Hungarian assignment + group association, with DBSCAN-based initiation, achieves the best overall occupancy-counting accuracy on `ee219_project2` (0.532), while maintaining low false positives (0.047 FP/frame) and the highest number of persistent tracks (6) in the reported comparison. Although some intermediate variants attain slightly higher occupancy recall or slightly lower FP/frame in isolation, the full saved pipeline provides the best overall tradeoff among stable tracking, counting accuracy, and duplicate- track suppression relative to simpler CFAR-only and 1-D Kalman baselines.

Remaining limitations are primarily rooted in the sensor's physical constraints: the 0.30 m range resolution prevents separation of two people in the same range bin, the dual-antenna baseline limits angular resolution to $\approx 51^\circ$ without super-resolution processing, and the 2-D EKF requires AoA measurements that are not always reliable at the current antenna configuration.

Future work should address three areas. First, improved AoA estimation using MUSIC [4] across the full captured sequence would enable 2-D trajectory reconstruction and reduce range-overlap ambiguity; the current PDoA/MVDR estimator is limited to $\approx 51^\circ$ angular resolution with only two antennas. Second, a probabilistic multi-hypothesis tracker (MHT) or joint probabilistic data association (JPDA) filter would better handle the ambiguous association events that remain challenging for the deterministic Hungarian pipeline. Third, real-time deployment on the STM32H743 — closing the loop between firmware CIR streaming and on-board tracking — would convert this offline research pipeline into an embedded occupancy sensor.

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